Rooftop Segmentation from Aerial Imagery: Cross-Dataset Transfer and Applicability



GROUP 10 Milestone 1

Course: DSAI Labs

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Term: September 2026

1. Problem Definition

1.1 Problem Statement

This project focuses on automatically detecting and segmenting building rooftops from high-resolution RGB aerial imagery. The task is to classify each pixel in an image as either rooftop or non-rooftop, producing a binary mask that shows the location and extent of rooftops. These masks can then be used to estimate rooftop area and support further applications, such as urban planning and solar energy assessment.

1.2 Primary Objective

The main objective of this project is to investigate how well standard rooftop segmentation models perform when applied to different datasets and to understand why their performance changes. Rather than developing a new model architecture or domain-adaptation method, the study uses a standard U-Net-style model as a fixed and well-understood baseline.

The focus is on examining how differences in geography, spatial resolution, image appearance, and label definition affect the model's ability to identify rooftops in unfamiliar data. Through controlled experiments, the project aims to provide a clearer understanding of the model's strengths, limitations, and suitability for use in new locations.

1.3 Primary Stakeholders

- Urban planners and municipal authorities: Need updated building maps to monitor urban growth, support city planning, and identify potentially unauthorized construction.
- Renewable energy planners and solar installers: Need reliable estimates of available rooftop space to identify potential locations for solar panel installation.

1.4 Secondary Stakeholders

- Disaster management agencies: Can use rooftop and building maps to support damage assessment, emergency response, and recovery planning.
- Drone navigation and logistics operators: May use rooftop information to identify potential landing surfaces and support aerial delivery planning.
- Defence and surveillance users: May use building and rooftop maps for geographic analysis and situational awareness.
- Researchers: Can use rooftop segmentation masks as input for further studies in remote sensing, urban analysis, and related applications.

1.5 Scope of the Project

The project will cover the following areas:

- Binary semantic segmentation of rooftops from RGB aerial or satellite imagery.
- Training and evaluation using three public benchmark datasets: Inria Aerial Image Labeling, AIRS, and Massachusetts Buildings.
- Evaluation under a consistent experimental protocol to ensure meaningful comparisons.
- Analysis of model performance across datasets with different spatial resolutions, geographic locations, image characteristics, and annotation styles.
- A preliminary evaluation using real-world imagery that was not included in the model's training data.

1.6 Limitations and Out-of-Scope Areas

The following tasks are outside the scope of this project:

- Individual building separation: Distinguishing and identifying each building as a separate instance.
- Roof classification: Identifying roof types, construction materials, or roof conditions.
- 3D reconstruction and height estimation: Recovering building heights or three-dimensional structures from imagery.
- Detailed solar analysis: Estimating solar irradiance, modelling shadows, assessing structural suitability, or designing solar panel layouts.
- Real-time deployment: Developing a system for real-time processing on mobile or embedded devices.
- New model development: Designing new segmentation architectures or developing domain-adaptation methods.

Overall, the project is designed to evaluate an existing rooftop segmentation approach rather than develop a new one. By keeping the model and evaluation procedure consistent, the study aims to identify the factors that influence transfer performance and establish how reliably the approach can be used with data from unfamiliar locations.

2. Why the Problem Matters and Where It Arises

Rooftop maps are an important source of information for many real-world decisions. However, creating these maps manually is slow, costly, and difficult to maintain at the city or national level. The problem is even greater in rapidly growing cities and developing regions, where satellite and aerial images are widely available, but labelled maps are often limited. In such cases, models trained in other regions may need to be reused, making transferability—how well a model works in a new area—a key challenge.

Renewable energy: Rooftop solar power is an important way to produce electricity locally. Large-scale programmes, such as India’s PM Surya Ghar rooftop solar scheme, need reliable estimates of suitable rooftops for solar panel installation. Tools like Google’s Project Sunroof show how automated rooftop analysis can support solar energy planning.

Urban governance: Comparing images taken at different times can help authorities identify unauthorized construction and track unplanned urban growth.

Disaster response: Updated building and rooftop maps can support quick damage assessments and help plan relief and rescue operations after disasters.

Drones and autonomous systems: Aerial images can help identify flat and safe surfaces for drone landings and deliveries. The Inria benchmark paper also highlights aerial imagery as a useful resource for applications such as autonomous driving and delivery.

Despite these benefits, identifying rooftops from aerial images is not easy. Roofs differ in shape, size, colour, and material, while shadows, seasonal changes, and differences between imaging sensors can affect how they appear. In densely populated areas, buildings often touch or overlap, making their boundaries difficult to identify. Trees, cars, and objects on rooftops add further confusion. Images taken from angled views create another challenge, as buildings may appear shifted from their actual ground positions due to relief displacement.

3. Literature Review: Existing Methods and Datasets

3.1 Methods and Datasets

3.1.1 Methods for Building and Rooftop Segmentation

In the early stages, building and rooftop detection mainly relied on hand-crafted features such as colour, texture, edges, and shape, combined with traditional classification methods. Researchers also used morphological techniques and object-based image analysis to identify buildings in aerial images. However, these approaches often struggled when applied to different locations because features that worked well in one region did not always perform equally well in another.

With the development of deep learning, pixel-wise segmentation became a practical way to identify buildings directly from images. Fully Convolutional Networks (Long et al., 2015) were among the first models to support this approach. Later, U-Net (Ronneberger et al., 2015) introduced an encoder-decoder structure with skip connections that help preserve fine spatial details. This made U-Net a widely used baseline for remote sensing image segmentation.

Further developments introduced other approaches to improve segmentation performance. Follow-up work by the Inria benchmark organizers reported promising results using U-Net-style architectures and combined loss functions. DeepLabv3+ (Chen et al., 2018) uses atrous convolutions to capture information at different scales, helping the model identify objects of varying sizes. Mask R-CNN (He et al., 2017) is useful when individual buildings need to be detected and separated rather than simply classified at the pixel level.

More recent approaches include SegFormer (Xie et al., 2021) and the Segment Anything Model (Kirillov et al., 2023). These methods offer alternative ways to understand and segment image content. However, their ability to perform reliably on aerial imagery without task-specific training remains an area of investigation.

3.1.2 Datasets for Building and Rooftop Segmentation

Several datasets have been developed to train and evaluate building and rooftop segmentation models. They differ in geographic coverage, spatial resolution, image characteristics, and the way building boundaries are labelled.

Massachusetts Buildings Dataset (Mnih, 2013): This is one of the early and widely used benchmarks for building detection, covering the Boston area. It provides a basis for evaluating segmentation methods on aerial imagery.

Inria Aerial Image Labeling Dataset (Maggiori et al., 2017): This dataset covers approximately 810 km² at a spatial resolution of 0.3 metres and contains two classes: building and non-building. Its training and test sets cover different cities, making it useful for evaluating how well models generalize to unfamiliar locations.

AIRS Dataset (Chen et al., 2019): AIRS provides high-resolution imagery at 7.5 cm per pixel and contains more than 220,000 buildings. It is particularly relevant to rooftop segmentation because it focuses on roof outlines. Its authors highlight that building footprints may not align with visible roof boundaries due to relief displacement in aerial imagery.

Other commonly used benchmarks include SpaceNet, the WHU Building Dataset (Ji et al., 2019), and DeepGlobe (2018). These datasets provide additional data for developing and evaluating building extraction methods across different settings.

Large-scale resources such as Microsoft Building Footprints and Google Open Buildings also demonstrate the potential of automated mapping over extensive geographic areas. However, they mainly provide building footprint data rather than roof-specific segmentation masks. This distinction matters when the goal is to identify visible rooftop areas accurately rather than the building's ground-level outline.

3.2 Synthesis: What Is Known and What Is Limited

3.2.1 Model Capability

What is known: Encoder-decoder convolutional neural networks (CNNs) have shown strong results in building and rooftop segmentation. This suggests that model architecture itself may not be the main barrier to practical use.

What is limited: Most reported improvements are based on data similar to that used during training. The performance of foundation models on aerial imagery is also still being studied.

What this project does: It uses a standard U-Net-style model as a fixed baseline so that performance differences can be linked more clearly to the data rather than changes in model architecture.

3.2.2 Generalization Across Datasets

What is known: Training and testing on different parts of the same image do not necessarily show whether a model can work well in a new location (Maggiori et al., 2017). Cross-dataset evaluations have reported major drops in performance. For example, Na et al. (2021) reported an IoU of 80.77 on the Inria validation set, compared with 48.46 on Massachusetts and 12.87 on WHU, when using a model trained on Inria.

What is limited: Although this performance drop has been documented, it is often treated as a general domain gap that needs to be addressed through domain adaptation. The specific reasons for the drop are not fully examined.

What this project does: It evaluates transfer across datasets using a complete transfer matrix and investigates which differences between datasets may explain the observed performance changes.

3.2.3 Label Definition

What is known: Building footprint labels do not always match actual roof outlines because of effects such as relief displacement (Chen et al., 2019). Many large-scale building maps also represent footprints rather than visible roof boundaries.

What is limited: Results from datasets labelled with building footprints and those labelled with roof outlines are not directly comparable. The effect of these differences on model performance has received limited attention, and AIRS was not included in the cross-dataset study discussed above.

What this project does: It includes AIRS alongside two building-labelled datasets and examines the types of errors made by the model, particularly errors along roof boundaries and cases where buildings are missed entirely.

3.2.4 Resolution and Geographic Differences

What is known: Existing datasets vary in spatial resolution, geographic location, imaging sensors, and image characteristics. For example, AIRS provides imagery at 7.5 cm resolution, while Inria uses 0.3 m resolution. Inria also includes test cities that are not present in its training data.

What is limited: These factors often change together across datasets, making it difficult to determine whether performance differences are caused by resolution, geography, sensor characteristics, or labelling practices.

What this project does: It attempts to examine these factors separately wherever possible. This includes resampling images to a common ground resolution and holding out cities within Inria to assess the effect of geographic differences.

3.2.5 Practical Use in Real-World Settings

What is known: Large-scale building footprint datasets demonstrate that automated mapping can be applied across extensive geographic areas.

What is limited: Real-world imagery often differs from carefully prepared benchmark datasets, yet models are rarely evaluated under these less controlled conditions.

What this project does: It adds an evaluation using imagery outside the selected benchmarks and reports performance changes to show how well the model transfers to more realistic settings.

3.3 What the Synthesis Implies

The reviewed studies consistently show that models trained on one dataset can perform much worse when applied to another. However, they provide less clarity on the specific reasons behind this decline. The cross-dataset study by Na et al. (2021), for example, proposes a domain-adaptation method to address the problem but treats the domain gap largely as a single issue rather than separating its possibe causes.

At the same time, existing datasets differ in spatial resolution, geographic location, sensor characteristics, and labelling conventions. Since these factors often vary together, it becomes difficult to identify which differences have the greatest effect on model performance.

As a result, someone planning to use a rooftop segmentation model in a new city or with a different imagery source has limited evidence to guide that decision. They may know that performance can decline, but not which factors are most responsible or how much accuracy to expect.

This project addresses that gap by evaluating cross-dataset transfer, examining the effects of dataset differences, and assessing how reliably a standard model can be applied to new locations and imagery sources. By keeping the model and evaluation procedure consistent, the study aims to provide a clearer understanding of when rooftop segmentation can be used reliably and where further testing or adaptation may be necessary.

4. Metrics and How Success Is Defined

Several evaluation metrics are used to measure how accurately a model identifies rooftops in aerial images. Each metric highlights a different aspect of model performance.

Intersection over Union (IoU / Jaccard Index): IoU measures the overlap between the predicted rooftop area and the actual rooftop area. It is calculated by dividing the intersection of the two areas by their union. It is the primary evaluation metric in many building segmentation benchmarks, including Inria.

Precision, Recall, and F1-Score (Dice): Precision measures how many predicted rooftop pixels are actually rooftops, while recall measures how many actual rooftop pixels the model successfully

identifies. The F1-score combines precision and recall into a single measure. These metrics are reported alongside IoU because rooftops usually occupy a smaller portion of an image than the background.

Pixel Accuracy: Pixel accuracy measures the proportion of all pixels classified correctly. However, it is reported only alongside IoU and F1-score because most pixels may belong to the background. As a result, a model can achieve high pixel accuracy while still missing many rooftops.

Relative Transfer Drop: This metric measures how much IoU decreases when a model is applied to a different dataset rather than the dataset it was trained on. It is calculated as:

Relative Transfer Drop (%) = [(In-Dataset IoU − Cross-Dataset IoU) / In-Dataset IoU] × 100

$$\text{Relative Transfer Drop} = \frac{\text{IoU}_{\text{in-dataset}} - \text{IoU}_{\text{cross-dataset}}}{\text{IoU}_{\text{in-dataset}}} \times 100$$

It expresses the performance decline as a percentage, making it easier to compare transfer results across dataset pairs with different starting performance levels.

Boundary-Aware Metrics: Metrics such as the Boundary F-score are used to examine how accurately predicted rooftop edges match the labelled boundaries. These metrics are particularly useful for studying differences between roof outlines and building footprints. Object-level metrics are not included because identifying and evaluating individual building instances is outside the scope of this study.

How Success Is Defined in This Study

In existing research, success is generally measured by achieving a higher IoU on a held-out test set from a public benchmark. However, this study does not aim simply to achieve the highest IoU score.

Instead, success means providing a clear, reproducible, and evidence-based understanding of how well a model transfers across datasets and why its performance changes. This includes identifying the effects of differences in resolution, geography, sensors, and label definitions, as well as reporting failures on data the model has never seen. The goal is to understand when rooftop segmentation can be trusted in a new setting and where its limitations remain.

5. Where Current Solutions Fall Short and the Opportunity

5.1 Key Research Gap

Previous studies have shown that building segmentation models often lose accuracy when applied to datasets different from those used during training. Researchers have also proposed methods to reduce this performance drop. However, the literature reviewed so far does not provide a controlled, like-for-like analysis of how individual dataset differences, such as spatial resolution, geography, and label definition, contribute to this loss. It also remains unclear how these factors affect datasets that use building footprints compared with those that show actual roof outlines.

5.2 Limitations of Existing Solutions

Label Inconsistency: Different datasets use different definitions of ground truth, including building footprints and roof outlines. They also vary in spatial resolution and annotation quality. As a result, performance scores across datasets are not always directly comparable, and the impact of these differences is rarely examined in detail.

Class Imbalance: Rooftops often occupy only a small part of an aerial image. Consequently, pixel accuracy can appear high even when a model misses smaller buildings or struggles to detect buildings in dense neighbourhoods.

Gap Between Benchmarks and Real-World Use: Real-world imagery, such as aerial data collected by government agencies or municipal authorities, may differ from benchmark datasets in image quality, resolution, and labelling standards. The performance of models on such imagery is rarely evaluated.

Resource Costs: Many recent segmentation models require considerable computing resources. There is still limited practical guidance on how well a lightweight, well-understood model can perform when applied to unfamiliar datasets or new locations.

5.3 Research Novelty and Opportunity

Cross-dataset evaluation itself is not new. Na et al. (2021), for example, evaluated transfer across the Inria, Massachusetts, and WHU datasets. Therefore, the novelty of this project lies in a more focused analysis of why transfer performance declines and how that decline can be understood.

The main contributions proposed in this study are:

- Identifying the causes of transfer loss: Examining the individual effects of dataset differences rather than treating them only as a general domain gap.
- Including different label definitions: Adding AIRS, which focuses on roof outlines, to the same transfer evaluation as datasets labelled with building footprints.
- Using a consistent evaluation protocol: Keeping the model and evaluation procedure fixed to make results more comparable and reproducible.

- Testing on real-world imagery: Adding an evaluation outside the selected benchmarks to examine how well the findings apply in practical settings.

Before Milestone 2, the team will conduct a systematic literature search covering cross-dataset evaluation, domain shift, generalization, and building and rooftop segmentation. This will help confirm the research gap and identify any relevant studies not covered in the current review.

The main opportunity is to carry out a systematic, like-for-like evaluation of a standard segmentation model across Inria, AIRS, and Massachusetts, followed by a check using real-world imagery. This will provide a clearer picture of how well rooftop detection transfers between datasets, which factors affect its performance, and where it is likely to fail. Such evidence can help practitioners decide when an existing model is suitable for a new location and when further adaptation or evaluation is needed.

6. Nature of the Contribution

6.1 Research Objective

The main objective of this project is to investigate how well a standard segmentation model transfers across rooftop datasets and to understand the reasons behind changes in its performance. The aim is to turn these findings into practical guidance for applying rooftop segmentation to unfamiliar locations and imagery sources.

The model itself is a tool for conducting this investigation rather than the main contribution of the research.

6.2 Research Questions

- RQ1: How much does a fixed baseline model lose in performance when trained on one of the Inria, AIRS, and Massachusetts datasets and tested on each of the others, using a consistent protocol to produce a 3 × 3 transfer matrix?

- RQ2: To what extent can the observed performance loss be linked to differences in spatial resolution, geography, image appearance, and label definition, particularly the distinction between roof outlines and building footprints?

- RQ3: To what extent do the results from the benchmark transfer matrix predict the model's performance on imagery collected outside these benchmarks?

6.3 Working Hypotheses

The following hypotheses will be tested during the study. They are provisional and may be rejected if the experimental results do not support them.

- H1: Cross-dataset IoU will be considerably lower than in-dataset IoU, and the size of the performance drop will vary depending on the direction of transfer between datasets.

- H2: Matching spatial resolution by resampling images to a common ground resolution will reduce a substantial part of the performance drop between datasets.

- H3: Differences in label definition will mainly produce errors around rooftop boundaries rather than errors caused by entire buildings being missed.

6.4 Expected Contribution

The main contribution of this project is an analysis of model robustness and generalization across different rooftop datasets. It is expected to provide four key outcomes:

1. Cross-dataset evaluation: A transfer matrix covering three datasets under a consistent experimental protocol.
2. Factor-level analysis: An investigation of how resolution, geography, image appearance, and label definition contribute to performance changes.
3. Practical guidance: Evidence showing what a lightweight baseline model can and cannot achieve when applied to unfamiliar imagery.
4. Real-world evaluation: An additional assessment using imagery outside the selected benchmarks to examine the practical relevance of the findings.

This project does not aim to introduce a new state-of-the-art segmentation architecture. Instead, it focuses on understanding the strengths and limitations of an existing approach under different conditions.

A secondary contribution is the practical use of the predicted rooftop masks. These masks can be used to estimate rooftop areas, providing a starting point for applications such as rooftop solar planning and other forms of urban analysis.

7. Evidence That Will Demonstrate the Approach Works

The success of this project will be evaluated using quantitative results, controlled experiments, visual comparisons, and testing on real-world imagery. The aim is not only to measure model accuracy but also to understand how reliably it performs across different datasets and where its limitations lie.

7.1 Quantitative Evaluation

The model will be evaluated using IoU, precision, recall, and F1-score on held-out test sets. Results will be reported for all nine train-test combinations across the Inria, AIRS, and Massachusetts datasets, including cases where the training and testing data come from the same dataset. This will allow a direct comparison of in-dataset performance and cross-dataset transfer. Pixel accuracy will not be used as the only measure of success.

7.2 Identifying the Causes of Performance Loss

Controlled experiments will be used to investigate why model performance changes across datasets. These will include matching spatial resolution through resampling and testing on held-out cities within the Inria dataset. Where the available data allow, these comparisons will help separate the effects of resolution and geography from other sources of variation.

7.3 Comparison with Existing Baselines

The results will be compared with published findings on the same benchmarks, where the evaluation settings are sufficiently comparable. A simpler alternative model will also be included to assess whether the selected baseline provides a reasonable level of performance. This comparison will help establish whether the observed transfer results reflect genuine challenges across datasets rather than weaknesses in the chosen model.

7.4 Qualitative Evaluation

Visual comparisons will be used to examine how well the model identifies rooftop boundaries and where its predictions differ from the ground-truth labels. Each comparison will include the original image, the ground truth, and the predicted mask. Particular attention will be given to difficult cases, such as roofs affected by shadows, densely packed buildings, and rooftops with unusual shapes or materials.

7.5 Evaluation on Real-World Imagery

The model will also be tested on imagery outside the selected benchmark datasets to assess how well it performs under more realistic conditions. The imagery source has not yet been finalised. Possible options include government or municipal aerial imagery, publicly available satellite imagery subject to its terms of use, and drone imagery captured by the project team.

The final source will be selected before Milestone 2 based on three criteria:

- Image quality: The imagery should provide RGB data at a spatial resolution reasonably comparable to the benchmark datasets.
- Permission for research use: Its terms of use must allow the intended research and analysis.
- Availability of labels: It should be possible to manually label a small sample of the imagery to support quantitative evaluation.

The results will include an honest assessment of any performance changes observed on this external imagery, including cases where the model struggles.

7.6 Reproducibility

All experiments will follow a consistent and documented procedure. Data splits will be fixed, preprocessing steps will be recorded, and the same evaluation code will be used across all datasets. These measures will make the results easier to verify, reproduce, and compare with future studies.

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